



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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GitHub Repository: <https://github.com/Whoikram/SpaceX-Landing-Prediction-Capstone>



Outline

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- Methodology
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Executive Summary

Summary of Methodologies

- Collected Falcon 9 launch data from the SpaceX API and Wikipedia.
- Performed data cleaning, feature engineering, SQL analysis, and exploratory data analysis.
- Developed interactive visualizations using Folium and Plotly Dash.
- Built and compared Logistic Regression, SVM, KNN, and Decision Tree models.
- Decision Tree achieved the highest prediction accuracy of 88.89%.

Model	Accuracy
Linear Regression	83.33%
SVM	83.33%
KNN	83.33%
Decision Tree	88.89%

Introduction

Introduction: SpaceX has significantly reduced launch costs by reusing the Falcon 9 first stage. Predicting whether the first stage will land successfully helps estimate launch costs and supports decision-making for organizations competing in the commercial space industry.

Project Objectives:

- Identify the factors affecting Falcon 9 first-stage landing success.
- Analyze launch data using SQL, visualization, and interactive dashboards.
- Build and compare machine learning models to predict landing outcomes.

Section 1

Methodology

Methodology

Executive Summary: This project employs a comprehensive approach to predict the successful landing of the Falcon 9 first stage, incorporating data collection, processing, exploratory analysis, interactive visualizations, and predictive modeling.

Data Collection Methodology: Data was sourced from the SpaceX API, which provided detailed records of Falcon 9 launches, including launch dates, sites, payloads, and outcomes.

Perform Data Wrangling: Data cleaning involved handling missing values, standardizing formats, and ensuring consistency. Key features were extracted and new features engineered to enrich the dataset.

Perform Exploratory Data Analysis (EDA) Using Visualization and SQL:

- Visualized launch success rates, payloads, and launch sites using Matplotlib and Seaborn.
- Executed SQL queries to derive insights and answer specific questions regarding the dataset.

Methodology

Perform Interactive Visual Analytics Using Folium and Plotly Dash:

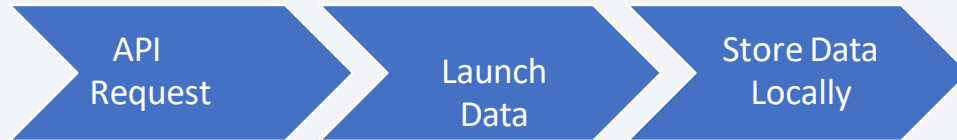
- Used Folium to create interactive maps displaying launch sites and outcomes.
- Developed a Plotly Dash application with interactive components like dropdowns and sliders to analyze launch success rates and payload ranges.

Perform Predictive Analysis Using Classification Models:

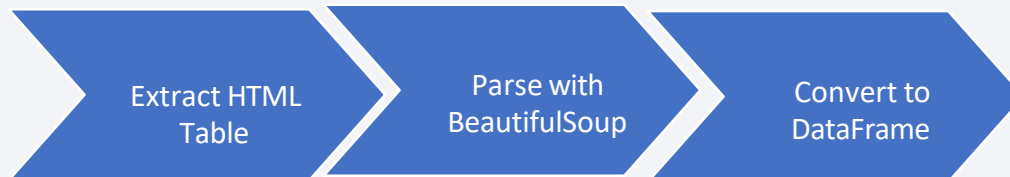
- Built and evaluated various classification models including Logistic Regression, SVM, KNN, and Decision Trees.
- Employed GridSearchCV for hyperparameter tuning.
- Evaluated models based on accuracy, and identified the best performing model for predicting landing success.

Data Collection

- Step 1: SpaceX API Request



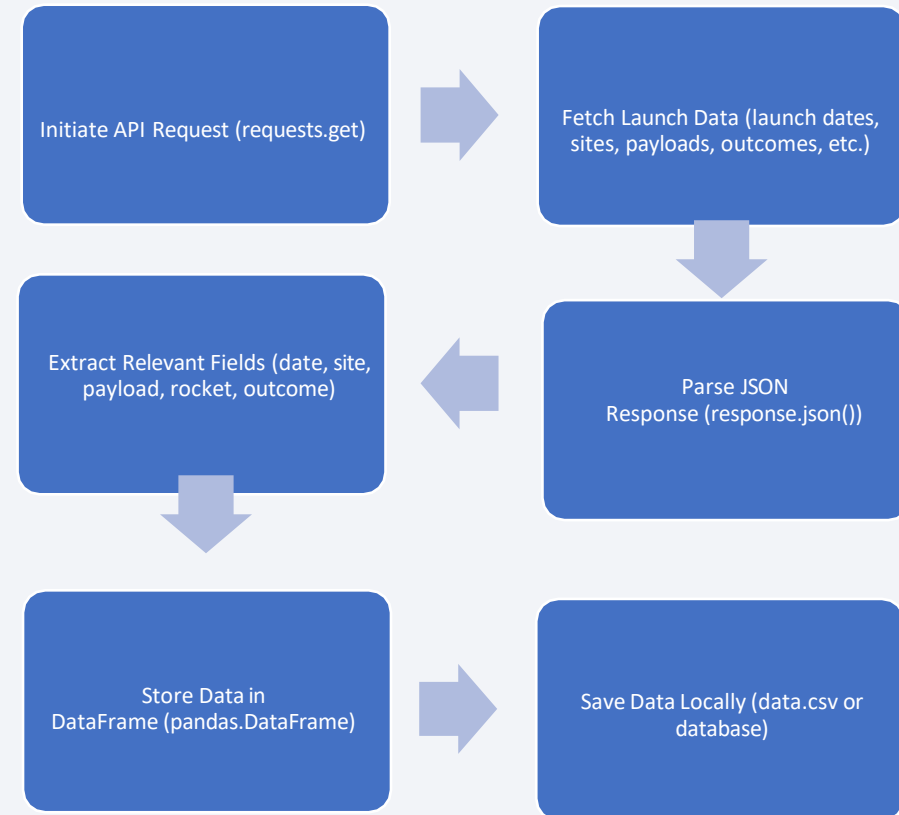
- Step 2: Web Scraping Wikipedia



Data Collection - SpaceX API

Data Collection Using SpaceX API

- Retrieved Falcon 9 launch data using the official SpaceX REST API.
- Extracted key attributes including Flight Number, Launch Site, Payload Mass, Booster Version and Landing Outcome.
- Stored the collected data in a Pandas Data Frame for further preprocessing and analysis.



GitHub URL 1 :<https://github.com/Whoikram/SpaceX-Landing-Prediction-Capstone/blob/main/1.%20jupyter-labs-spacex-data-collection-api.ipynb>

Data Collection - Web Scrapping

Step 1: Initiate Web Scrapping

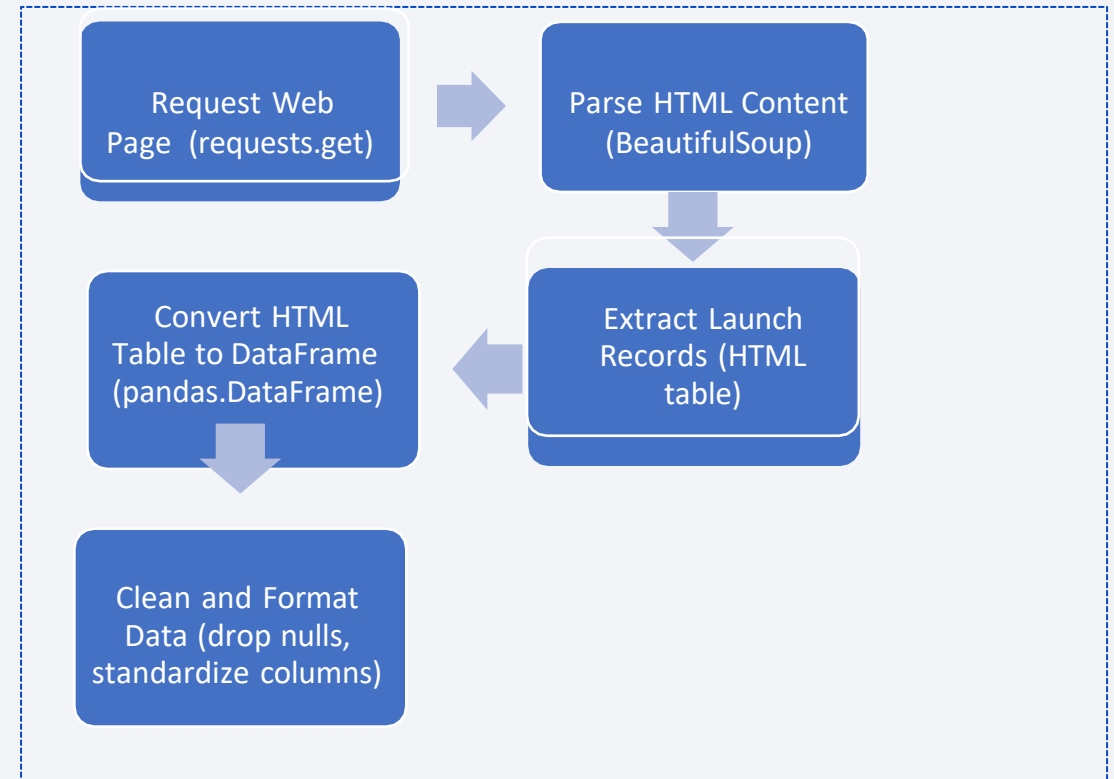
- Use Python's `requests` library to fetch the HTML content of the Wikipedia page.
- Target URL:
[https://en.wikipedia.org/wiki/List of Falcon 9 and Falcon Heavy launches](https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)

Step 2: Parse HTML Content

- Use `BeautifulSoup` to parse the HTML content.
- Extract the HTML table containing Falcon 9 launch records.

Step 3: Convert to DataFrame

- Convert the extracted HTML table into a pandas DataFrame.
- Clean and format the DataFrame, ensuring data consistency.



GitHub URL 2:<https://github.com/Whoikram/SpaceX-Landing-Prediction-Capstone/blob/main/2.%20jupyter-labs-webscraping.ipynb>

Data Wrangling

Overview: Data wrangling involves transforming, and organizing raw data into a structured format suitable for analysis.

- Step 1: Data Cleaning
 - Identify and fill or remove missing values in the dataset.
 - Use appropriate imputation techniques such as filled with mean, median or removing null values directly
- Step 2: Data Transformation
 - Convert data types to appropriate formats (e.g., date-time, numerical).
 - Standardize text (e.g., lowercase, remove whitespace).
 - Create new features from existing data (e.g., extract year from date).
 - Use Standard Scaler numerical features to ensure consistency.

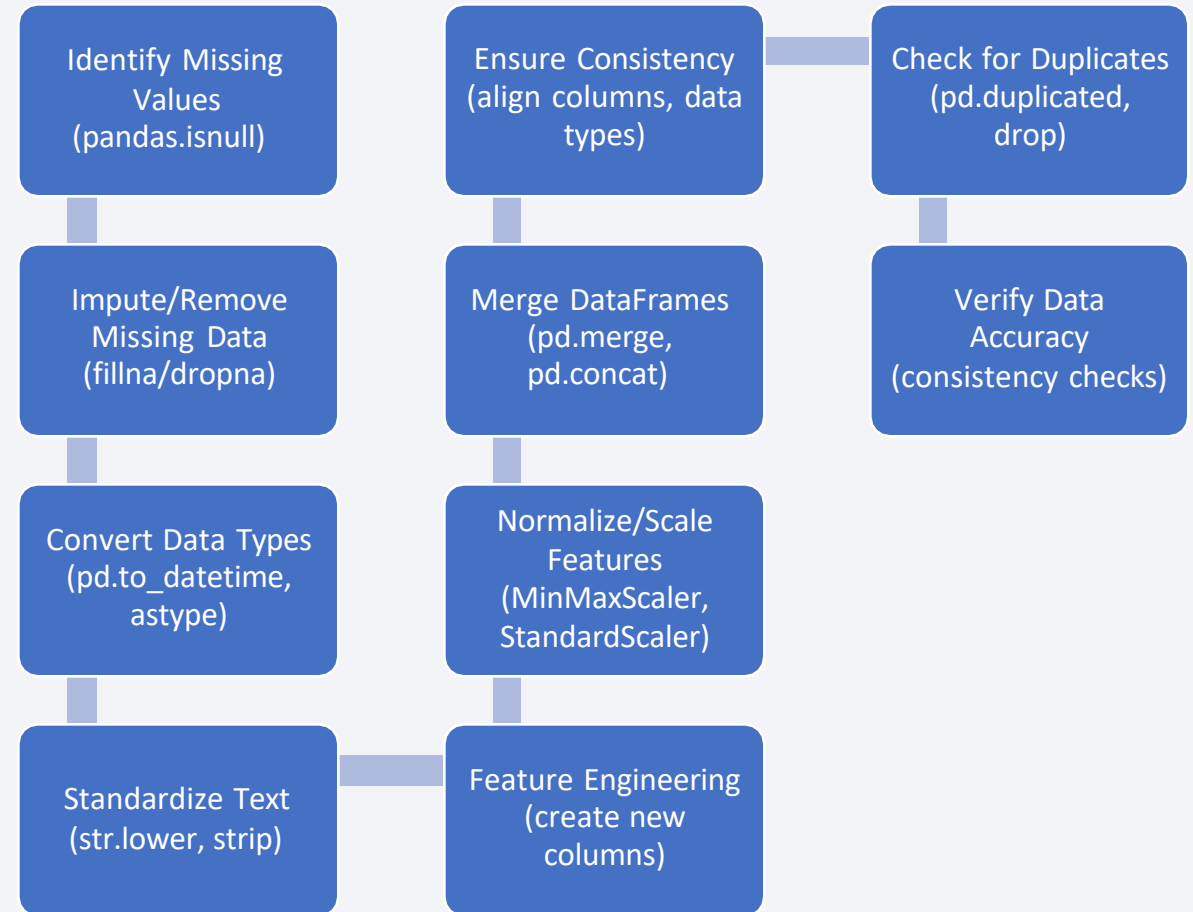
GitHub URL 3:<https://github.com/Whoikram/SpaceX-Landing-Prediction-Capstone/blob/main/3.%20labs-jupyter-spacex-Data%20wrangling.ipynb>

Data Wrangling

- Step 3: Data Integration
 - Merge datasets collected from different sources (API, web scraping) into a single cohesive dataset.
 - Ensure consistent column names and data formats across datasets.
- Step 4: Data Validation
 - Check for duplicate records and remove them.
 - Verify the accuracy and consistency of data entries.

Data Wrangling Flowchart

This flowchart summarizes the complete data wrangling workflow used in this project. The raw data collected from the SpaceX API and Wikipedia was cleaned by handling missing values, converting data types, standardizing text, engineering new features, removing duplicates, and validating data consistency. These preprocessing steps ensured a high-quality dataset for exploratory data analysis and machine learning.



EDA with Data Visualization

Overview:

Exploratory Data Analysis (EDA) involves visually exploring and summarizing the main characteristics of a dataset. The goal is to understand the data's distribution, identify patterns, and uncover relationships between variables.

Charts Plotted:

1. Histograms:

- **Purpose:** Used to visualize the distribution of numerical variables such as launch success rates, payload mass, and flight number.
- **Why:** Helps in understanding the spread and central tendency of the data, identifying outliers, and assessing data skewness.

2. Bar Charts:

- **Purpose:** Used to compare categorical variables such as launch outcomes (success/failure) across different categories like launch sites or rocket types.
- **Why:** Provides a clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends.

3. Line Charts:

- **Purpose:** Used to track trends over time, such as the success rate of Falcon 9 launches across different years.
- **Why:** Reveals temporal patterns and helps in understanding performance trends or changes over specific periods.

EDA with Data Visualization

4. Scatter Plots:

- **Purpose:** Used to explore relationships between two numerical variables, such as payload mass vs. launch success.
- **Why:** Identifies correlations or dependencies between variables, visualizing how one variable changes concerning another.

5. Heatmaps:

- **Purpose:** Used to visualize correlation matrices between multiple numerical variables.
- **Why:** Helps in identifying strong correlations (positive or negative) between variables, aiding feature selection or understanding multicollinearity.

6. Box Plots:

- **Purpose:** Used to display the distribution of numerical data through their quartiles.
- **Why:** Visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories.

EDA with SQL

Aggregate Queries:

- Calculated total number of launches.
- Counted successful and failed launches.
- Calculated success rates by launch site and rocket type.

Filtering Queries:

- Filtered data to focus on specific launch outcomes (success/failure).
- Applied conditions to extract launches based on criteria like launch date or rocket configuration.

Sorting Queries:

- Sorted data to identify trends or outliers.
- Ordered launches by date or success rate for analysis.

Subqueries:

- Nested queries to calculate derived metrics (e.g., average payload mass per launch site).
- Subqueries used to perform detailed analysis within larger datasets.

Build an Interactive Map with Folium

Map Objects Created

Markers:

- Placed markers to indicate launch sites on the map.
- Each marker represents a specific geographical location where SpaceX launches have occurred.

Circles:

- Added circles around launch sites to visually represent proximity zones.
- Circles help visualize the areas around launch sites that might influence operational decisions.

Lines:

- Drew lines to connect launch sites with their proximities or other relevant locations.
- Lines provide spatial context and connections between different points of interest related to launches.

Reasons for Adding Objects

Markers:

- To pinpoint exact launch locations for spatial reference.
- Helps users identify where SpaceX has conducted launches geographically.

Circles:

- Illustrates the potential impact zones around launch sites.
- Provides a visual representation of safety perimeters or operational boundaries.

Lines:

- Shows connections or relationships between launch sites and relevant features.
- Enhances understanding of spatial relationships and dependencies.

GitHub URL6: https://github.com/Whoikram/SpaceX-Landing-Prediction-Capstone/blob/main/6.%20lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Plots/Graphs Added

Success Pie Chart:

- Displays the distribution of successful and failed launches.
- Helps visualize the overall success rate and performance trends.

Success-Payload Scatter Plot:

- Shows the relationship between payload mass and launch success.
- Allows users to explore how payload mass influences mission outcomes.

Interactions Added

Launch Site Dropdown:

- Enables users to select specific launch sites for analysis.
- Facilitates filtering and focused exploration based on geographical locations.

Range Slider for Payload:

- Allows users to adjust payload mass ranges dynamically.
- Offers flexibility in examining launch success concerning payload mass variations.

Reasons for Adding Plots and Interactions

Success Pie Chart:

- Provides a quick overview of mission success rates.
- Essential for stakeholders to understand overall performance metrics at a glance.

Success-Payload Scatter Plot:

- Helps identify correlations between payload characteristics and launch outcomes.
- Supports decision-making processes related to payload planning and operational strategies.

Launch Site Dropdown:

- Enhances user experience by focusing analysis on specific launch locations.
- Allows for regional insights and comparisons across different launch sites.

Range Slider for Payload:

- Offers interactive exploration of how payload mass affects mission success.
- Enables detailed analysis and insights into payload-related performance factors.

Predictive Analysis (Classification)

1. Data Preprocessing:

- Standardized features to ensure all variables contribute equally.
- Split data into training and test sets for model validation.

2. Model Selection:

- Explored multiple classification algorithms: SVM, Decision Trees, and K-Nearest Neighbors (KNN).
- Chose algorithms suitable for binary classification tasks based on project requirements.

3. Hyperparameter Tuning:

- Used GridSearchCV to systematically search for optimal hyperparameters.
- Tuned parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (KNN).

4. Model Evaluation:

- Evaluated models using cross-validation techniques to ensure robustness and generalizability.
- Utilized metrics like accuracy, precision, recall, and F1-score to assess model performance.

5. Improvement Iterations:

- Iteratively adjusted models based on insights from validation results.
- Fine-tuned hyperparameters to maximize predictive accuracy and reliability.

6. Selection of Best Performing Model:

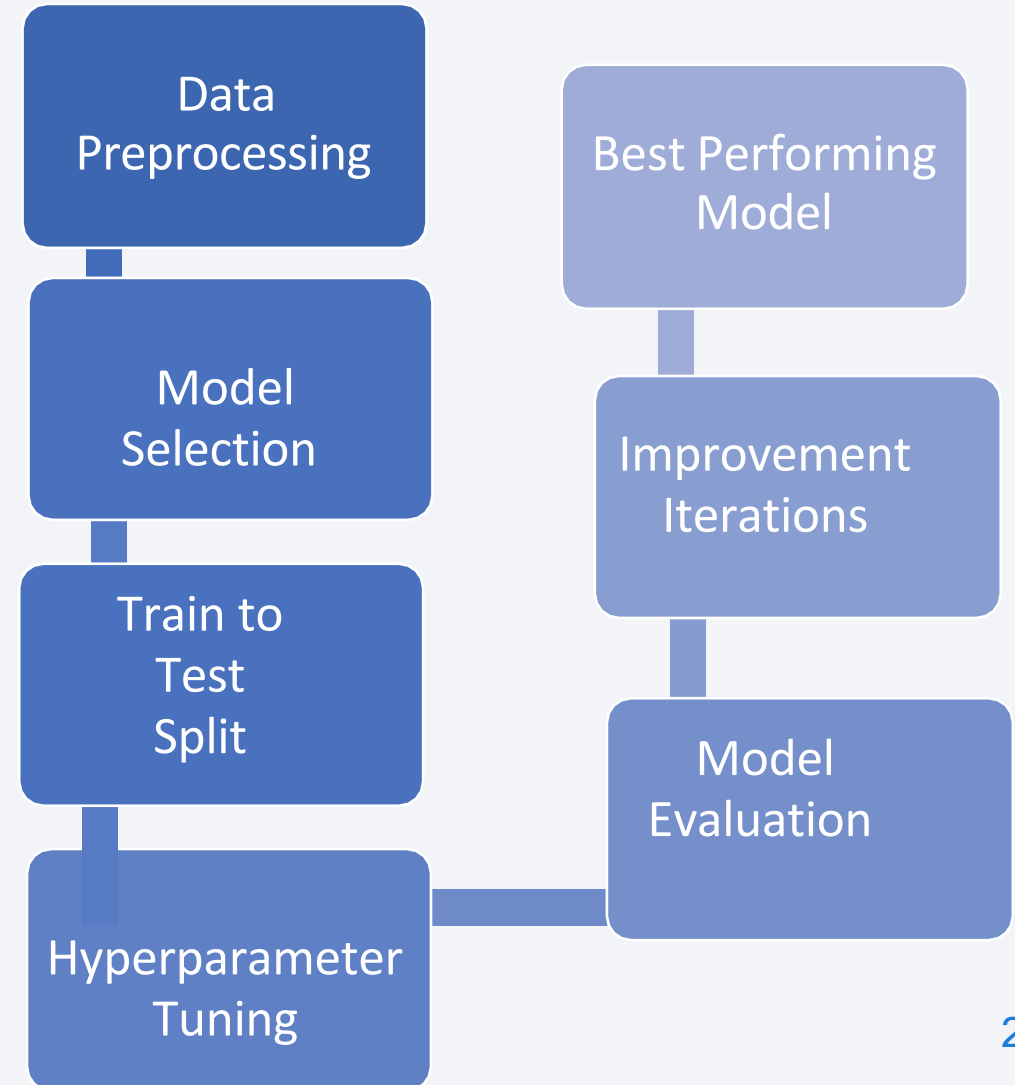
- Identified the model with the highest accuracy on the test set as the best performer.
- Considered both training and test set performance to avoid overfitting and ensure real-world applicability.

GitHub URL7: https://github.com/Whoikram/SpaceX-Landing-Prediction-Capstone/blob/main/7.%20SpaceX_Machine_Learning_Prediction_Part_5.jupyterlite.ipynb

Predictive Analysis (Flowchart)

Machine Learning Workflow:

This flowchart summarizes the complete classification pipeline used in the project. The processed data was split into training and testing sets, multiple classification models were trained and optimized using GridSearchCV, and the best-performing model was selected based on test accuracy. This workflow ensured reliable and unbiased prediction of Falcon 9 first-stage landing success.

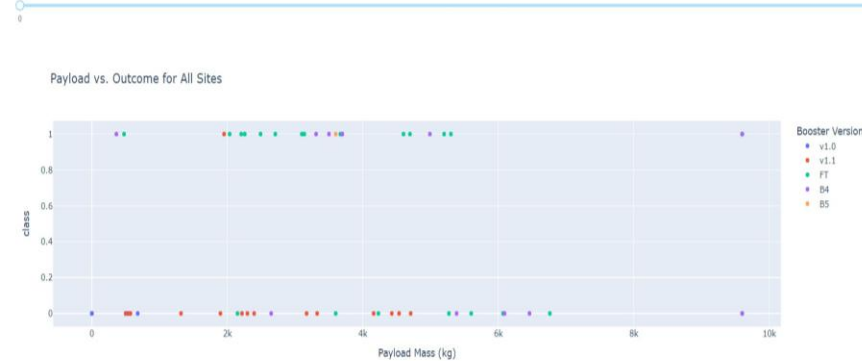


All Sites

Total Success Launches by Site



Payload range (Kg):



Results

TASK 12

Find the method performs best:

Based on the results I got:

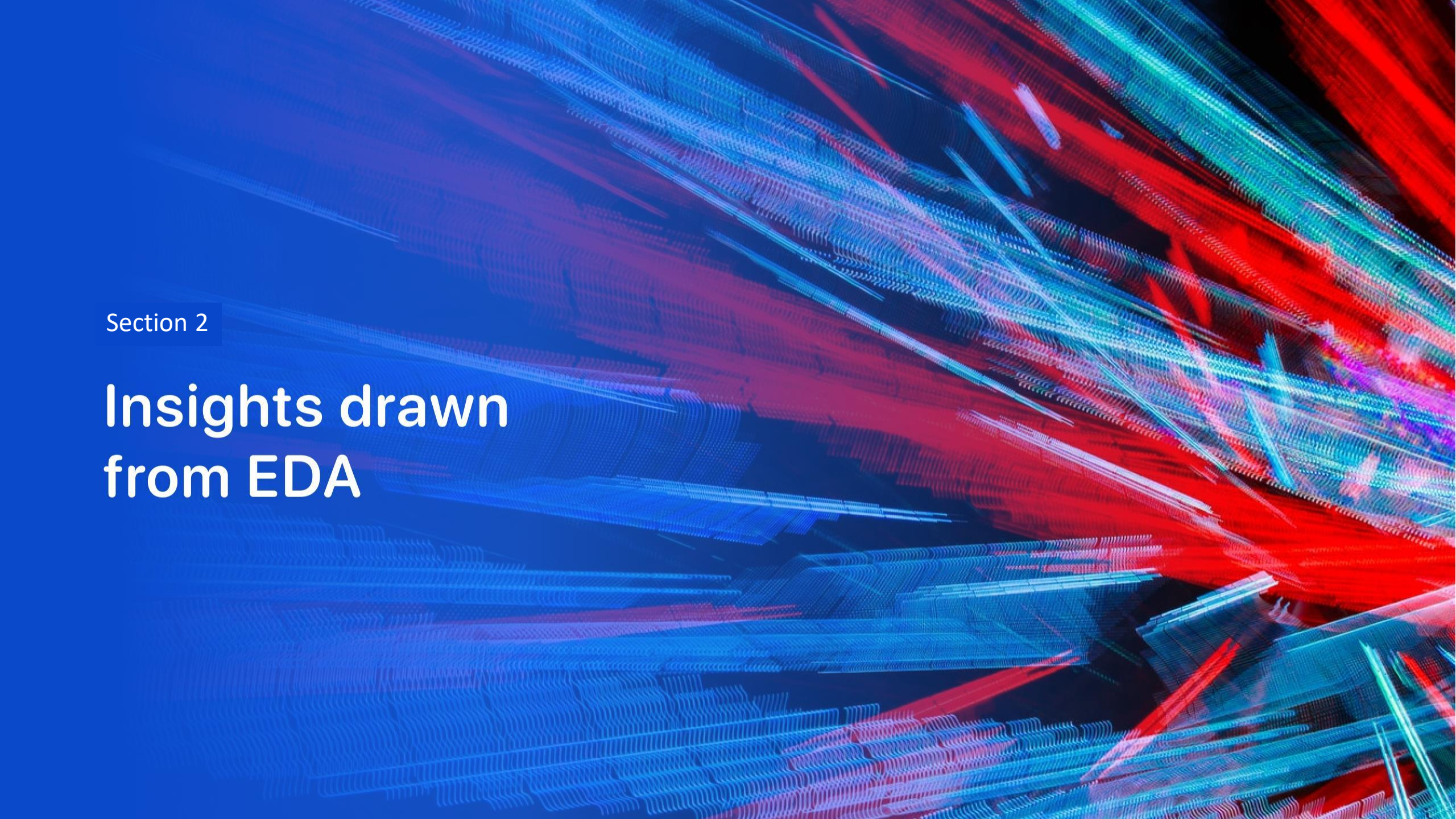
Logistic Regression: Accuracy on test data: 0.8333, Support Vector Machine (SVM): Accuracy on test data: 0.8333, Decision Tree: Accuracy on test data: 0.8888, K Nearest Neighbors (KNN): Accuracy on test data: 0.8333

It seems there was a significant increase in accuracy for the Decision Tree model, which shows the highest accuracy of 0.8888 on the test set. Therefore, in this comparison, the Decision Tree model performs the best among the classifiers you've trained.

Authors

Pratiksha Verma

- **Classifier Evaluation: > Best Performer:** The Decision Tree model achieved the highest predictive performance with an outstanding test accuracy of 88.88%.
- **Alternative Models:** Logistic Regression, SVM, and KNN models all tied at a lower test data accuracy of 83.33%.

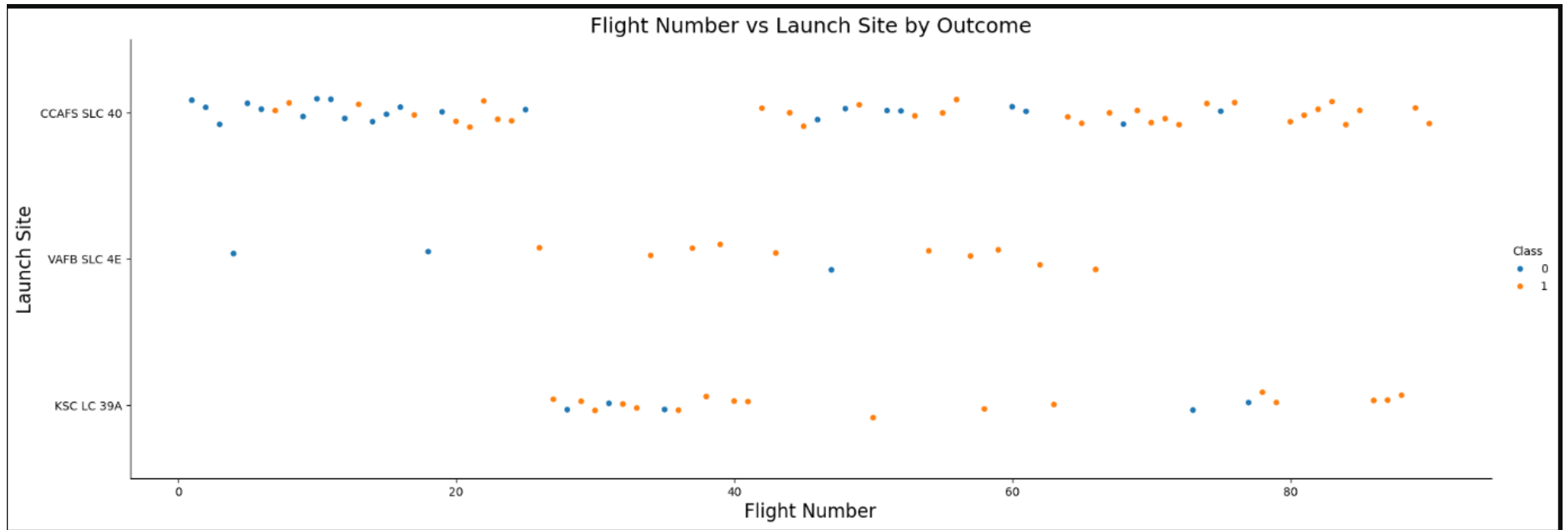


Section 2

Insights drawn from EDA

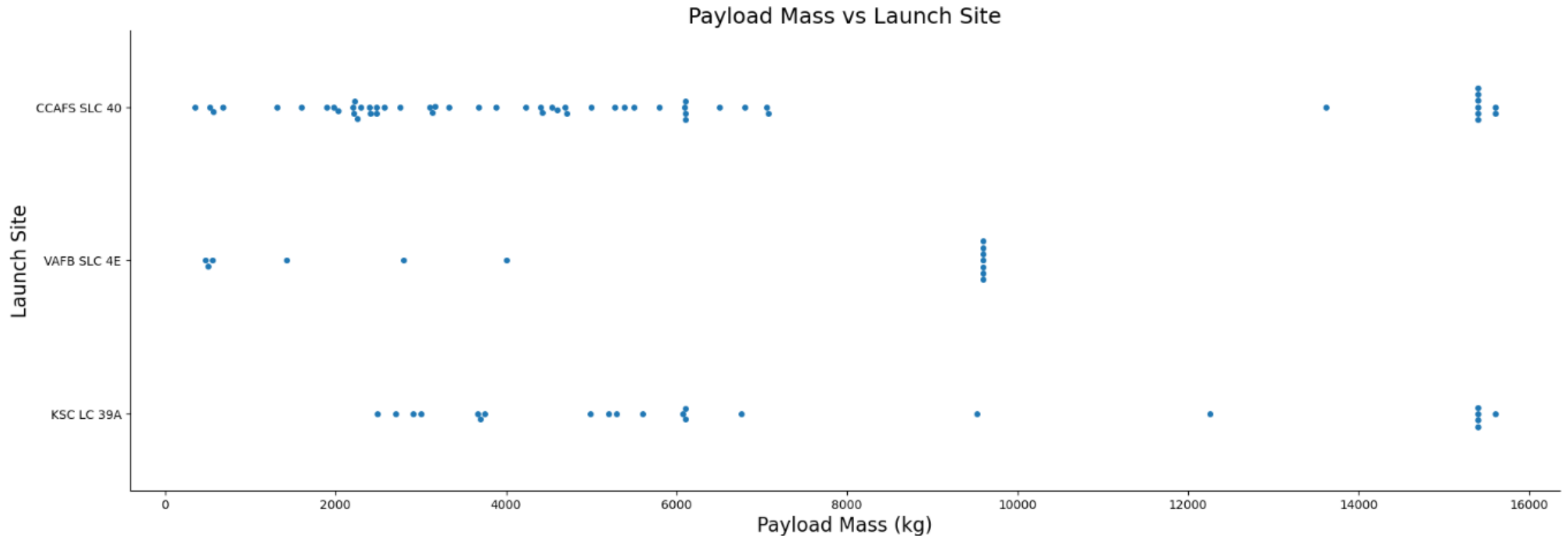
Flight Number vs. Launch Site

- **Mixed Outcomes at Major Launch Sites:** Both CCAFS SLC 40 and KSC LC 39A have a mix of successful (orange) and unsuccessful (blue) landings, indicating that factors other than the launch site itself may influence the landing success.
- **Consistent Activity Across Flight Numbers:** Launches are spread across a wide range of flight numbers at all sites, suggesting consistent activity over time without a clear trend of increasing or decreasing landing success.



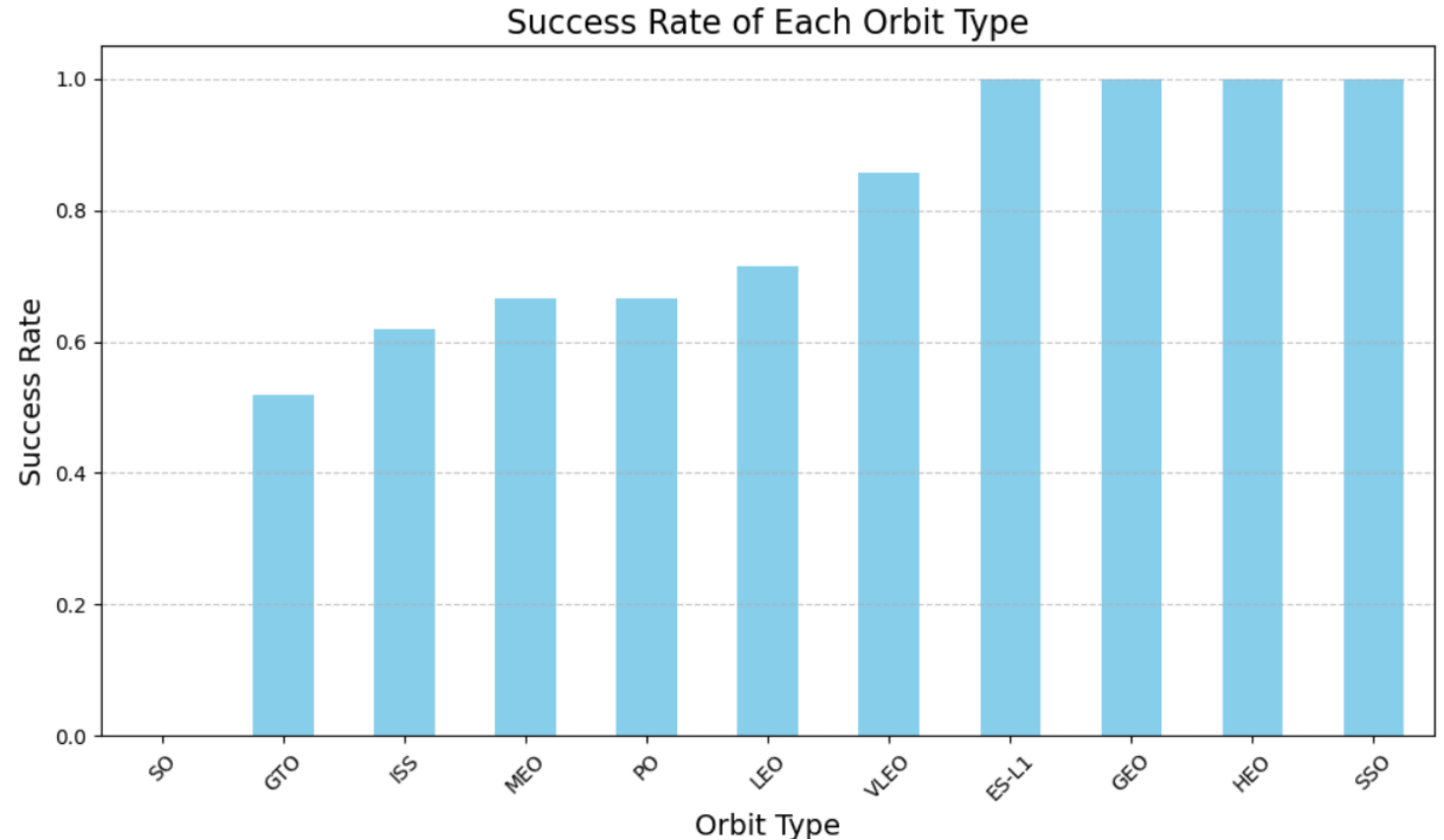
Payload vs. Launch Site

- **Payload Distribution:** Most launches from the CCAFS SLC 40 site handle payloads below 10,000 kg, while the VAFB SLC 4E and KSC LC 39A sites have a wider range of payload masses, indicating varied mission profiles.
- **High-Capacity Launches:** The KSC LC 39A site is frequently used for launching heavier payloads, with multiple launches carrying over 15,000 kg, suggesting its suitability for high-capacity missions.



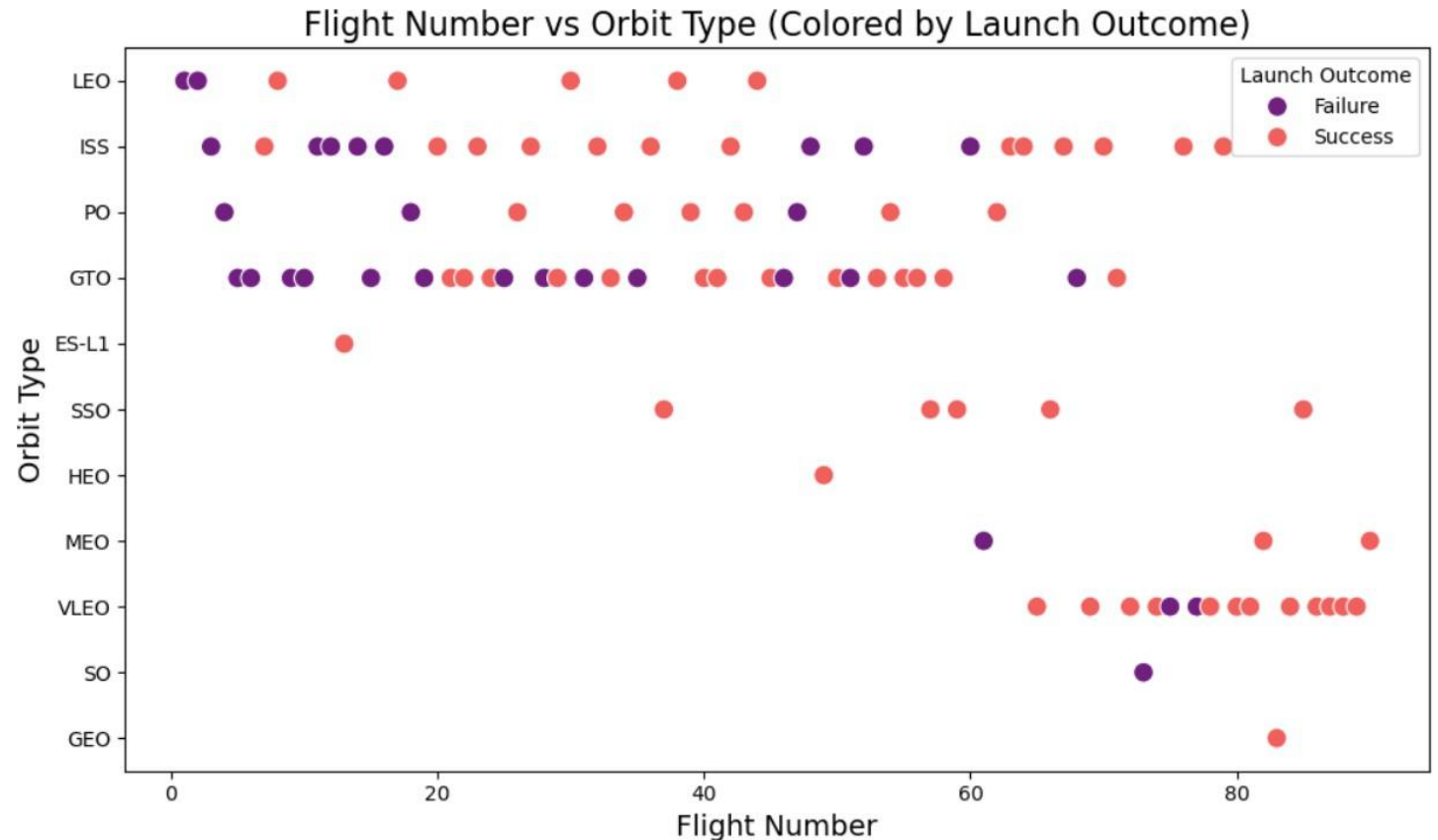
Success Rate vs. Orbit Type

- **High Success Rates:** Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits have achieved a perfect success rate, indicating these orbits are highly reliable for successful first stage landings.
- **Lower Success Rate for GTO:** The GTO orbit type shows a significantly lower success rate compared to other orbit types, suggesting that missions to this orbit may involve greater challenges or complexities.



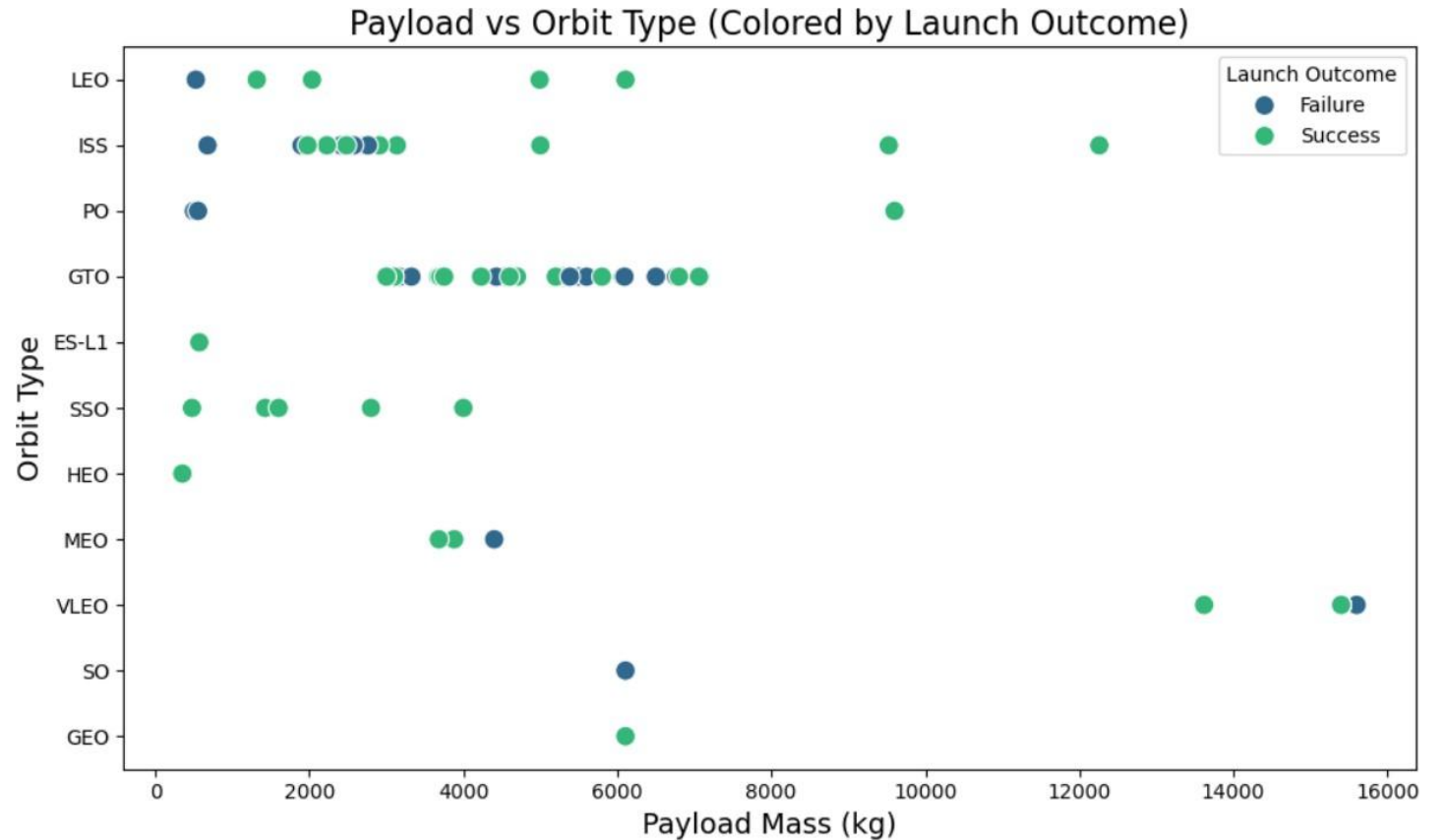
Flight Number vs. Orbit Type

- **Increased Success Over Time:** The success rate of Falcon 9 launches improves significantly with higher flight numbers, indicating that experience and iterative improvements contribute to better outcomes.
- **Orbit-Specific Performance:** Early flights to GTO and ISS orbits had mixed outcomes, but recent missions to these orbits show a higher success rate, reflecting advancements in mission planning and execution.



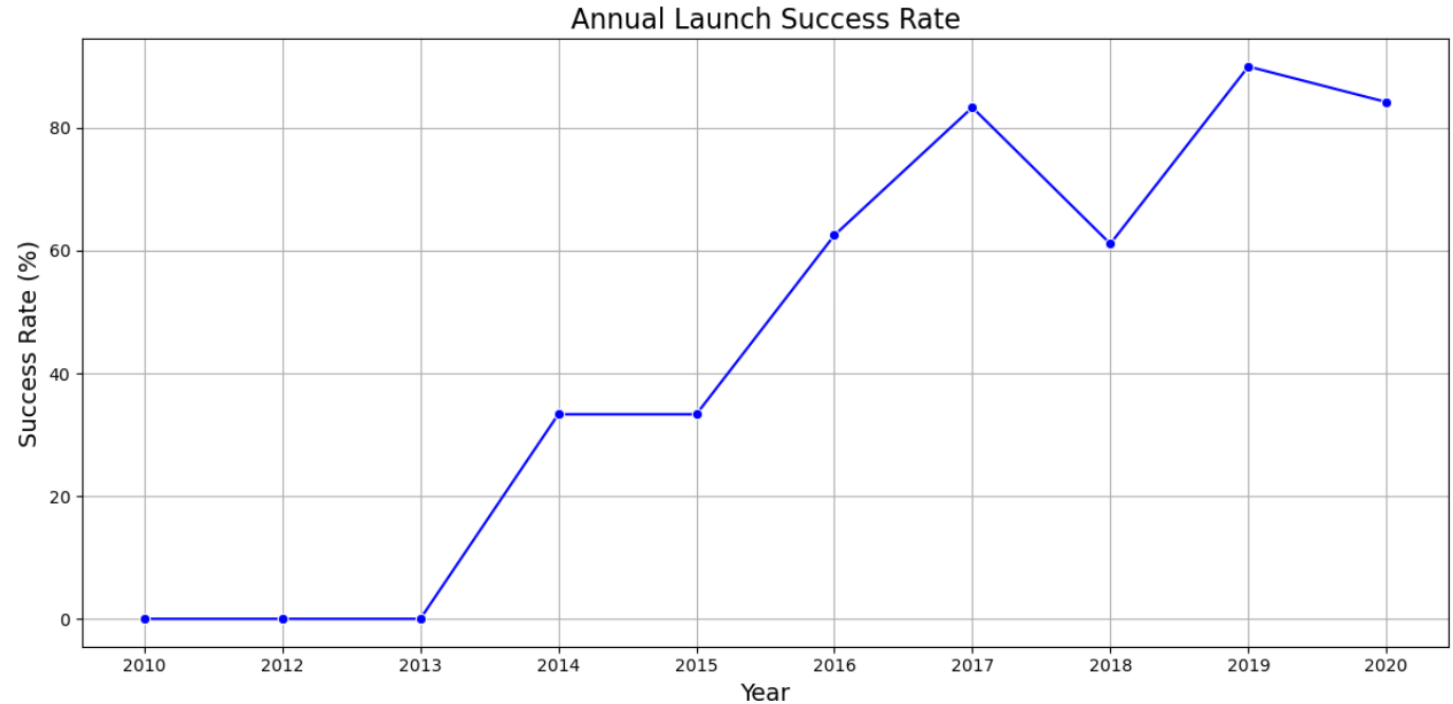
Payload vs. Orbit Type

- Successful landings are more frequent across all orbit types, especially for payloads less than 6000 kg.
- Higher payload masses (above 10,000 kg) show a mix of successes and failures, indicating increased difficulty with heavier payloads.



Launch Success Yearly Trend

- The annual launch success rate has shown a significant improvement from 2013 onwards, reaching over 80% by 2020.
- Despite a dip in 2018, the overall trend indicates increasing reliability and success in Falcon 9 launches over the years.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[21]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

```
[21]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

The Falcon 9 dataset contains **four unique launch sites**, representing the primary locations used by SpaceX for satellite deployment and resupply missions. Identifying unique launch sites helps analyze how launch location influences payload characteristics and first-stage landing success.

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[26]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

```
[26]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

This query demonstrates how pattern matching can be used to efficiently filter launch records. It also shows that multiple Falcon 9 missions were conducted from Cape Canaveral launch facilities, highlighting their importance in SpaceX operations.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[30]: %sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[30]: SUM(PAYLOAD_MASS_KG_)
```

```
45596
```

The total payload mass carried by all booster rockets launched for the customer *NASA (CRS)* sums up to exactly **45,596 kg**.

This result highlights the significant role of Falcon 9 in supporting NASA cargo resupply missions to the International Space Station. Calculating the total payload mass provides insight into the scale of cargo transported and demonstrates the usefulness of SQL aggregate functions for mission analysis.

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[34]: %sql SELECT AVG("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[34]: AVG(PAYLOAD_MASS_KG_)
```

```
2928.4
```

The Average Payload Mass carried by all booster rockets launched for booster version F9 v1.1 is **2,928.4 kg**

This result provides an overview of the carrying capacity of the Falcon 9 v1.1 booster. Calculating the average payload mass helps compare booster performance and supports analysis of how payload characteristics may influence mission outcomes and landing success..

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[36]: %sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';  
* sqlite:///my_data1.db  
Done.
```

```
[36]: MIN(Date)  
-----  
2015-12-22
```

The first successful Falcon 9 ground-pad landing occurred on **December 22, 2015**.

This milestone marked a major breakthrough in reusable rocket technology. Successfully landing the first stage on a ground pad demonstrated SpaceX's ability to recover and reuse boosters, helping reduce launch costs and improve mission efficiency.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[38]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000
```

```
* sqlite:///my_data1.db  
Done.
```

```
[38]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

The boosters **F9 FT B1022, B1026, B1021.2, and B1031.2** successfully completed drone ship landings while carrying payloads between **4,000 kg and 6,000 kg**

This result demonstrates that Falcon 9 successfully recovered its first stage even while carrying medium-weight payloads. It highlights the reliability of drone ship recovery operations and shows how SQL filtering can identify missions meeting specific payload and landing conditions.

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[40]: %sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[40]:
```

Mission_Outcome	Total
Success	98

The SQL query shows that the Falcon 9 dataset contains **98 successful mission outcomes**, along with the corresponding failed mission outcomes. This provides an overall summary of mission performance across all recorded launches.

Boosters Carried Maximum Payload

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
[42]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);  
* sqlite:///my_data1.db  
Done.
```

```
[42]: Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

The SQL query identified **F9 B5 B1048, B1049, and B1051** variants as the booster cores that carried the **maximum payload mass** among all Falcon 9 missions.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[69]: %sql
SELECT
  CASE
    WHEN substr("Date", 6, 2) = '01' THEN 'January'
    WHEN substr("Date", 6, 2) = '02' THEN 'February'
    WHEN substr("Date", 6, 2) = '03' THEN 'March'
    WHEN substr("Date", 6, 2) = '04' THEN 'April'
    WHEN substr("Date", 6, 2) = '05' THEN 'May'
    WHEN substr("Date", 6, 2) = '06' THEN 'June'
    WHEN substr("Date", 6, 2) = '07' THEN 'July'
    WHEN substr("Date", 6, 2) = '08' THEN 'August'
    WHEN substr("Date", 6, 2) = '09' THEN 'September'
    WHEN substr("Date", 6, 2) = '10' THEN 'October'
    WHEN substr("Date", 6, 2) = '11' THEN 'November'
    WHEN substr("Date", 6, 2) = '12' THEN 'December'
    ELSE 'Unknown'
  END AS "Month_Name",
  "Mission_Outcome",
  "Booster_Version",
  "Launch_Site"
FROM
  SPACEXTABLE
WHERE
  substr("Date", 0, 5) = '2015';

* sqlite:///my_data1.db
Done.
```

```
[69]:
```

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

The SQL query retrieved Falcon 9 launch records for **2015**, showing that **CCAFS LC-40** experienced mostly successful launches, with only **one in-flight failure** recorded during the year.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[81]: %%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;
```

```
* sqlite:///my_data1.db
Done.
```

```
[81]:
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

The SQL query ranked Falcon 9 landing outcomes recorded between **June 4, 2010** and **March 20, 2017**. During this period, **"No Attempt"** was the most common landing outcome, followed by successful and failed drone ship landings.

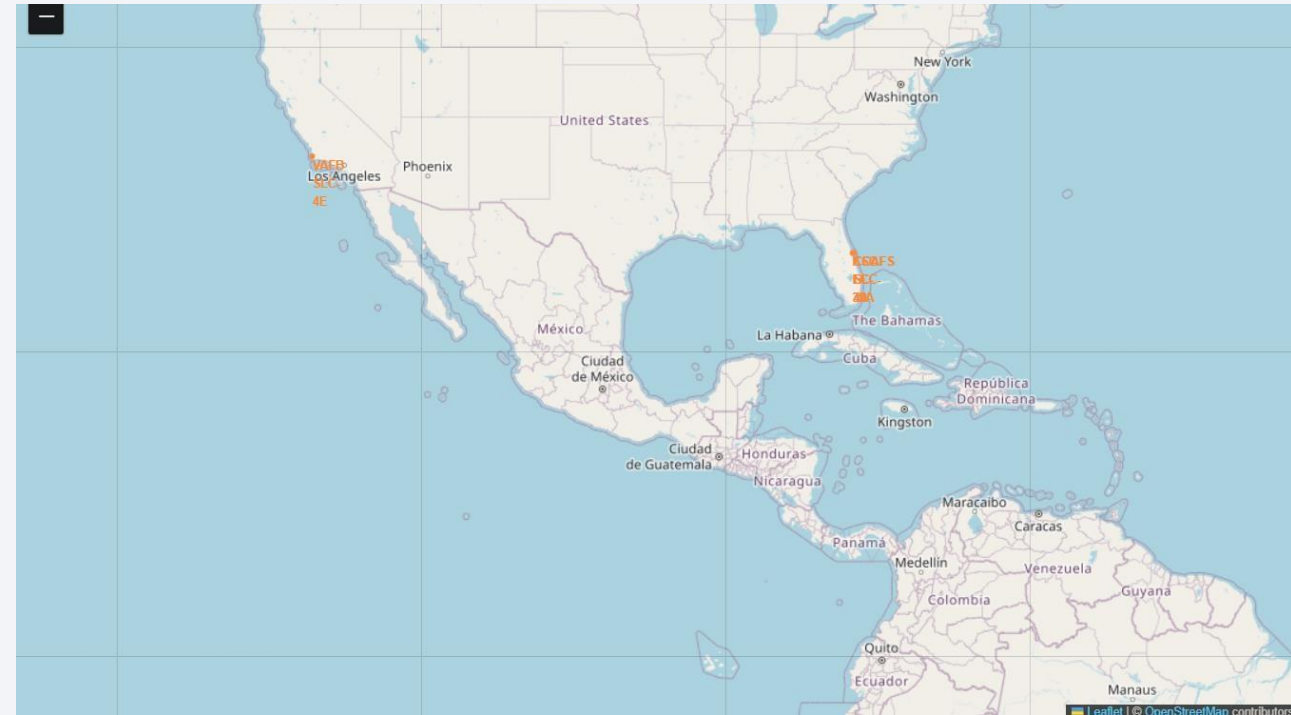
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue rectangle on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible, separating the dark surface from the deep blue of the sky.

Section 3

Launch Sites Proximities Analysis

Task 1: Mark all launch sites on a map

1. The Folium map displays the geographical locations of all four Falcon 9 launch sites. The visualization shows that the Florida launch sites (**CCAFS LC-40**, **CCAFS SLC-40**, and **KSC LC-39A**) are located closer to the Equator than **VAFB SLC-4E** in California. Additionally, all launch sites are positioned near the coastline, providing safe over-water launch trajectories and reducing potential risks to populated areas.
2. The geographic distribution of launch sites demonstrates how launch location influences mission planning. Sites closer to the Equator can benefit from the Earth's rotational speed, while coastal locations improve operational safety by allowing launch vehicles to travel over the ocean during ascent.



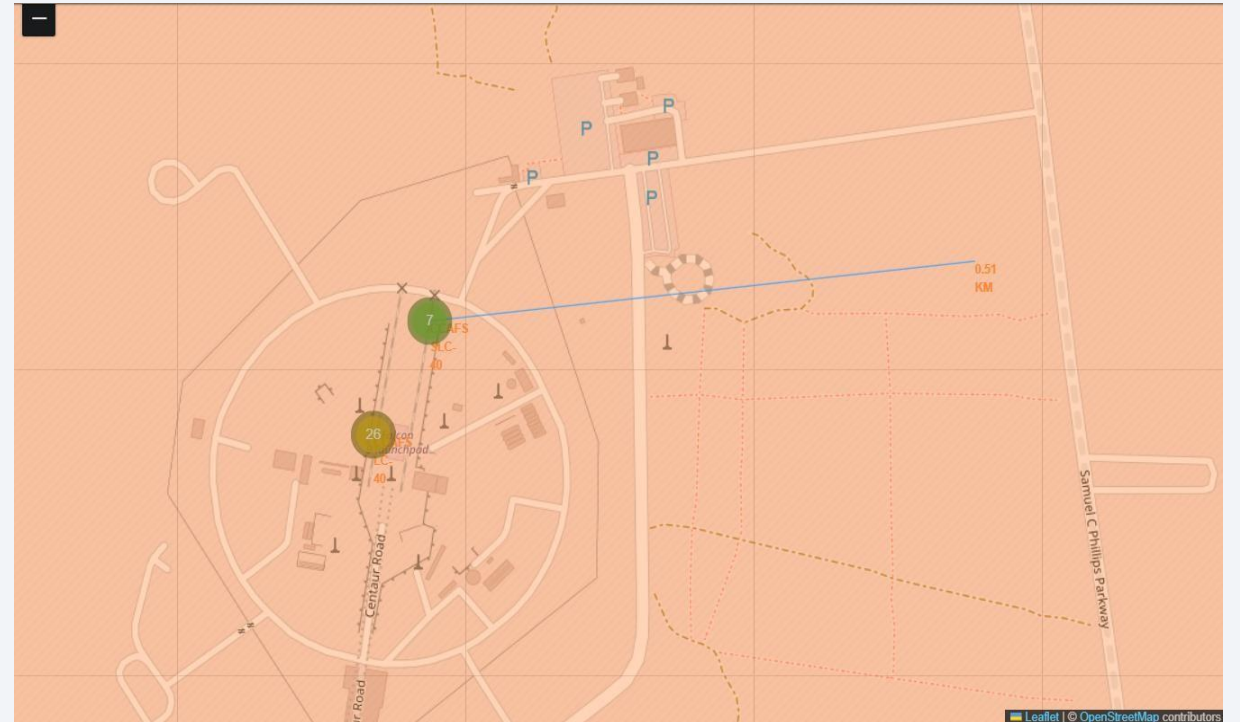
Task 2: Mark the success/failed launches for each site on the map

- **Map Analysis:** This Folium visualization displays all Falcon 9 launch locations using clustered markers, making it easier to explore mission data without overlapping markers. The color-coded markers distinguish successful and unsuccessful launches, while pop-up information provides additional mission details. The clustering feature improves readability and helps identify launch activity patterns across different sites.
- **Key Insight:** The map reveals that launch outcomes vary across different launch sites, with some locations exhibiting a higher concentration of successful missions. Interactive clustering enables efficient exploration of mission performance and provides valuable geographical insights into Falcon 9 launch operations.



Task 3: Calculate the distances between a launch site to its proximities

Map Analysis: This Folium visualization measures the distance between the **CCAFS SLC-40** launch site and the nearest coastline. The calculated distance is approximately **0.51 km**, and the PolyLine illustrates the shortest path between the two locations. This analysis demonstrates the use of geospatial calculations to evaluate the physical characteristics of launch sites.



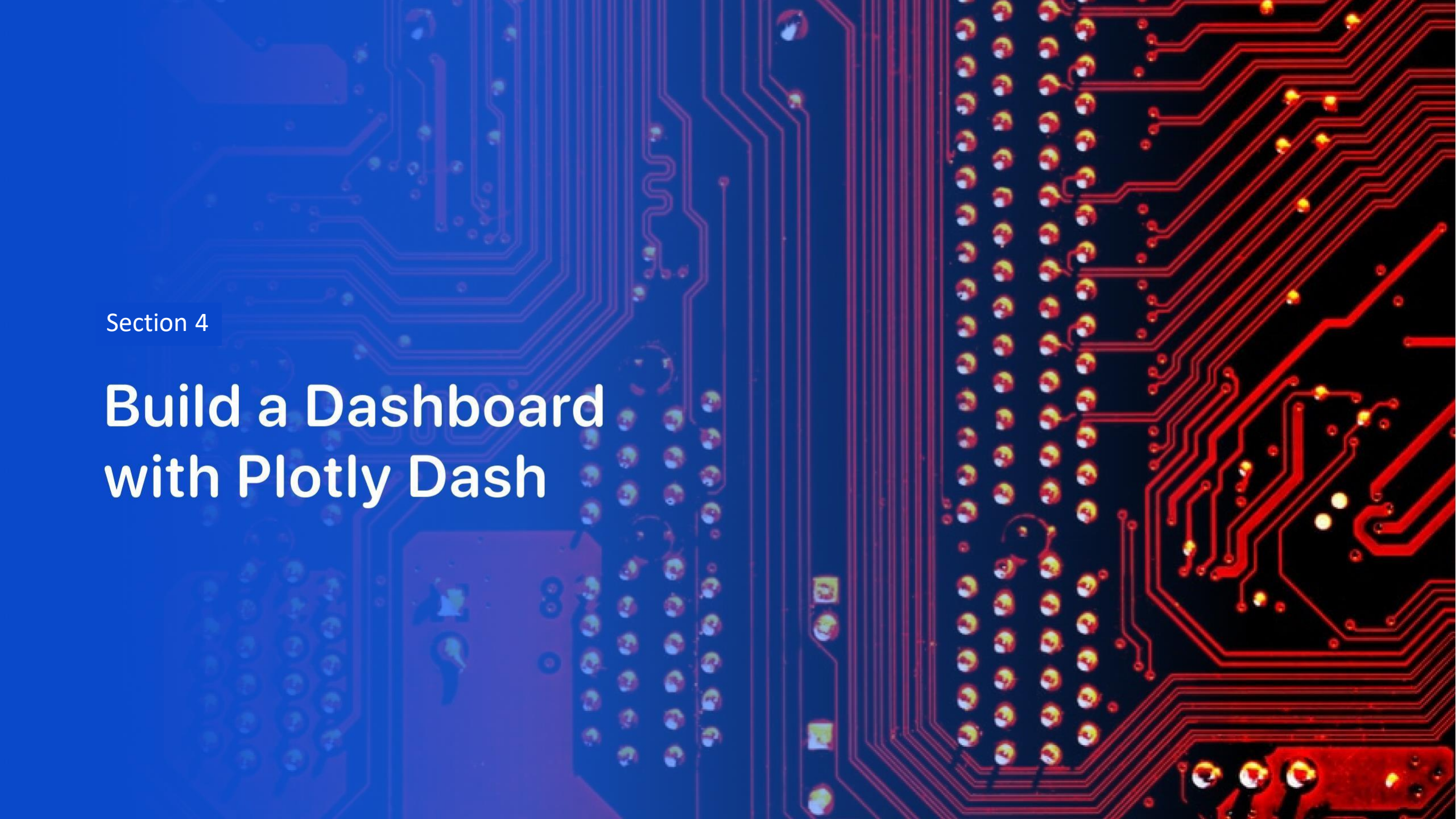
```
[43]: coastline_lat = 28.56367
      coastline_lon = -80.57163

      # Example launch site coordinates (replace with actual launch site coordinates)
      launch_site_lat = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Lat'].values[0]
      launch_site_lon = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Long'].values[0]

      # Calculate distance using the calculate_distance function
      distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)

      print(f"Distance from launch site to closest coastline: {distance_coastline} km")

      Distance from launch site to closest coastline: 0.5097439631188213 km
```

Section 4

Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

All Sites

Total Success Launches by Site



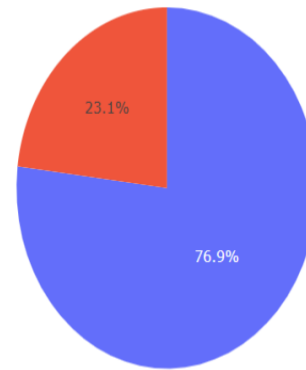
Payload range (Kg):



Dashboard Analysis: The pie chart illustrates the distribution of successful Falcon 9 launches across all launch sites. **KSC LC-39A** contributes the highest proportion of successful launches (**41.7%**), followed by **CCAFS LC-40 (29.2%)**, **VAFB SLC-4E (16.7%)**, and **CCAFS SLC-40 (12.5%)**

Key Insight: The dashboard reveals that **KSC LC-39A** has the largest share of successful launches, indicating its strong operational performance and reliability. Comparing launch success across different sites helps identify which locations consistently achieve better mission outcomes and supports data-driven decision-making.

Total Success Launches for site KSC LC-39A



Dashboard Analysis: The pie chart displays the launch outcome distribution for **KSC LC-39A**, the launch site with the highest success ratio. Approximately **76.9%** of launches were successful, while **23.1%** resulted in unsuccessful outcomes, demonstrating the site's consistently strong performance.

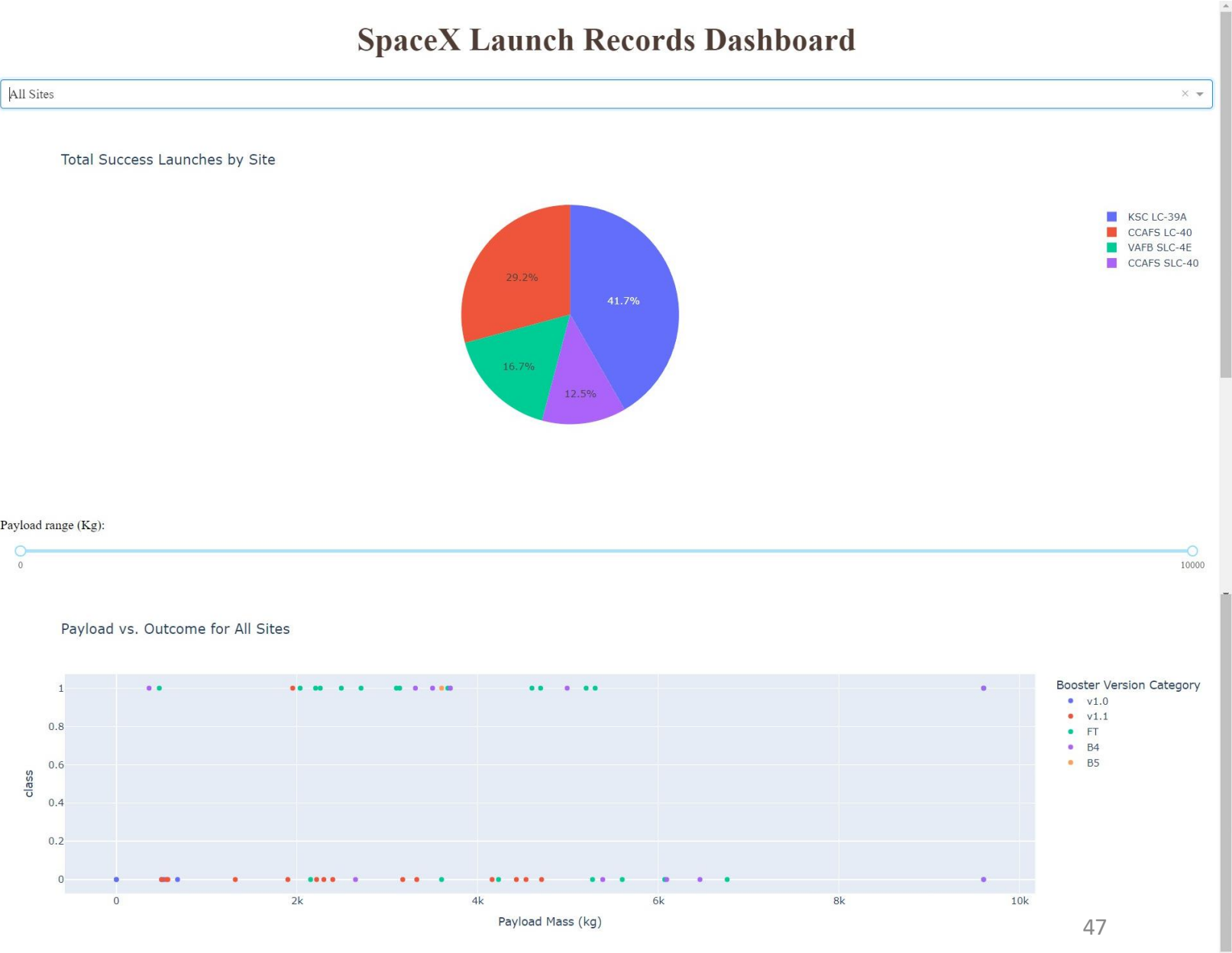
Key Insight: The high success ratio at **KSC LC-39A** indicates that this launch site has been one of SpaceX's most reliable operational facilities. Analyzing launch outcomes at individual sites helps identify performance differences and supports strategic planning for future missions.

Key Insights from SpaceX

Launch Data Dashboard

Dashboard Summary:
Launch Site Performance: The dashboard shows that launch success varies across different launch sites. **CCAFS LC-40** demonstrates consistently strong mission performance, while the remaining launch sites exhibit varying success rates depending on mission characteristics and payload requirements.

Booster Version Analysis: The **FT** booster version appears most frequently and achieves consistently successful launches across a wide range of payload masses. Older booster versions such as **v1.0** have fewer recorded missions, making performance comparisons less representative.



Section 5

Predictive Analysis (Classification)

Classification Accuracy

- **Model Comparison:** Four machine learning models were evaluated for predicting Falcon 9 first-stage landing success. The **Decision Tree** model achieved the highest test accuracy of **88.89%**, outperforming **Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)**, each of which achieved **83.33%** accuracy.

```
[51]: print("tuned hpyerparameters :(best parameters) ",tree_cv.best_params_)  
      print("accuracy :",tree_cv.best_score_)
```

```
tuned hpyerparameters :(best parameters) {'criterion': 'gini', 'max_depth': 16, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 5, 'splitter': 'best'}  
accuracy : 0.8732142857142857
```

TASK 9

Calculate the accuracy of tree_cv on the test data using the method `score`:

```
[52]: tree_cv.score(X_test,Y_test)
```

```
[52]: 0.8888888888888888
```

We can plot the confusion matrix

Confusion Matrix

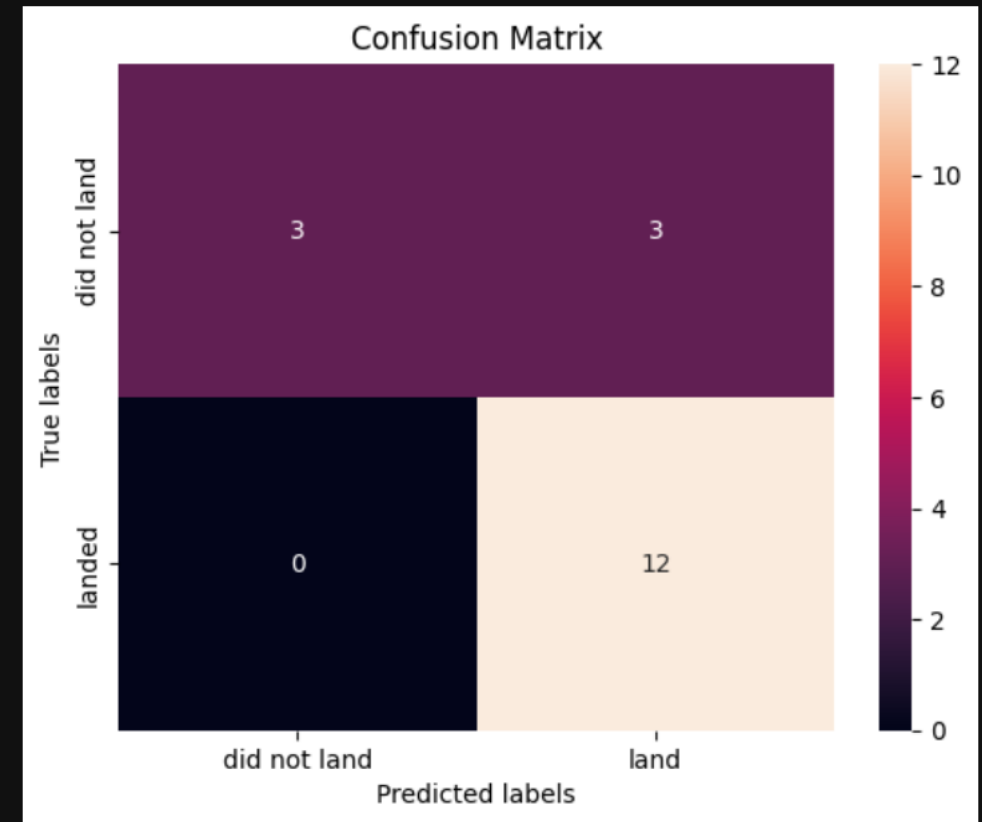
Explanation and Insights:

The confusion matrix evaluates the performance of the **Decision Tree classifier** in predicting Falcon 9 first-stage landing outcomes. The model correctly classified **12 successful landings (True Positives)** and **3 unsuccessful landings (True Negatives)**. It produced **3 False Positives**, where unsuccessful landings were incorrectly predicted as successful, and **0 False Negatives**, meaning every successful landing was correctly identified.

The model achieved an overall classification accuracy of **88.89%**, demonstrating strong predictive performance. The absence of false negatives indicates excellent reliability in identifying successful landings, while the small number of false positives has minimal impact on the overall model performance.

Key Insight: The confusion matrix confirms that the Decision Tree model provides accurate and dependable predictions, making it the most suitable classifier for predicting Falcon 9 first-stage landing success in this project.

```
[41]: yhat = tree_cv.predict(X_test)
      plot_confusion_matrix(Y_test,yhat)
```



Conclusions

Key Findings

1. Launch Site Performance

The analysis revealed that launch success varies across different launch sites. **KSC LC-39A** and **CCAFS LC-40** consistently demonstrated strong operational performance and high mission success rates.

2. Booster Version Performance

The **Falcon 9 FT** booster version achieved consistently successful launches across a wide range of payload masses, highlighting its reliability and operational maturity.

3. Payload Analysis

No strong relationship was observed between higher payload mass and lower landing success, indicating that mission outcome depends on multiple operational factors rather than payload mass alone.

4. Interactive Visual Analytics

Folium maps and Plotly Dash dashboards provided valuable geographical and operational insights, enabling interactive exploration of launch sites, payload characteristics, and mission outcomes.

Exploratory Data Analysis successfully identified the key operational factors that influence Falcon 9 first-stage landing success and provided meaningful insights for predictive modeling.

Conclusions

Project Conclusion

This project successfully combined **data collection, data wrangling, SQL analysis, exploratory data analysis, interactive visualizations, and machine learning** to investigate the factors affecting Falcon 9 first-stage landing success.

Among the evaluated models, the **Decision Tree classifier** achieved the highest prediction accuracy (**88.89%**), making it the most effective model for this dataset.

The analysis demonstrated that **launch site characteristics, booster version, and mission-specific features** have a greater influence on landing success than payload mass alone.

Overall, this project demonstrates how **data science and machine learning** can support decision-making in aerospace applications by improving launch planning, reducing operational costs, and enabling more accurate prediction of reusable rocket landings.

Thank you!

